

TetraCraft is a universe built from first principles.

This is an architecture for unified molecular physics. A “kleiniverse” is modeled from the interaction of fundamental tetrahedral units, or TETs.

These building blocks bond, react, and evolve into complex chemical structures governed by magnetic polarity and energy exchange.

It is a digital playground for exploring the intersection of matter, consciousness, and the genesis of complex systems.

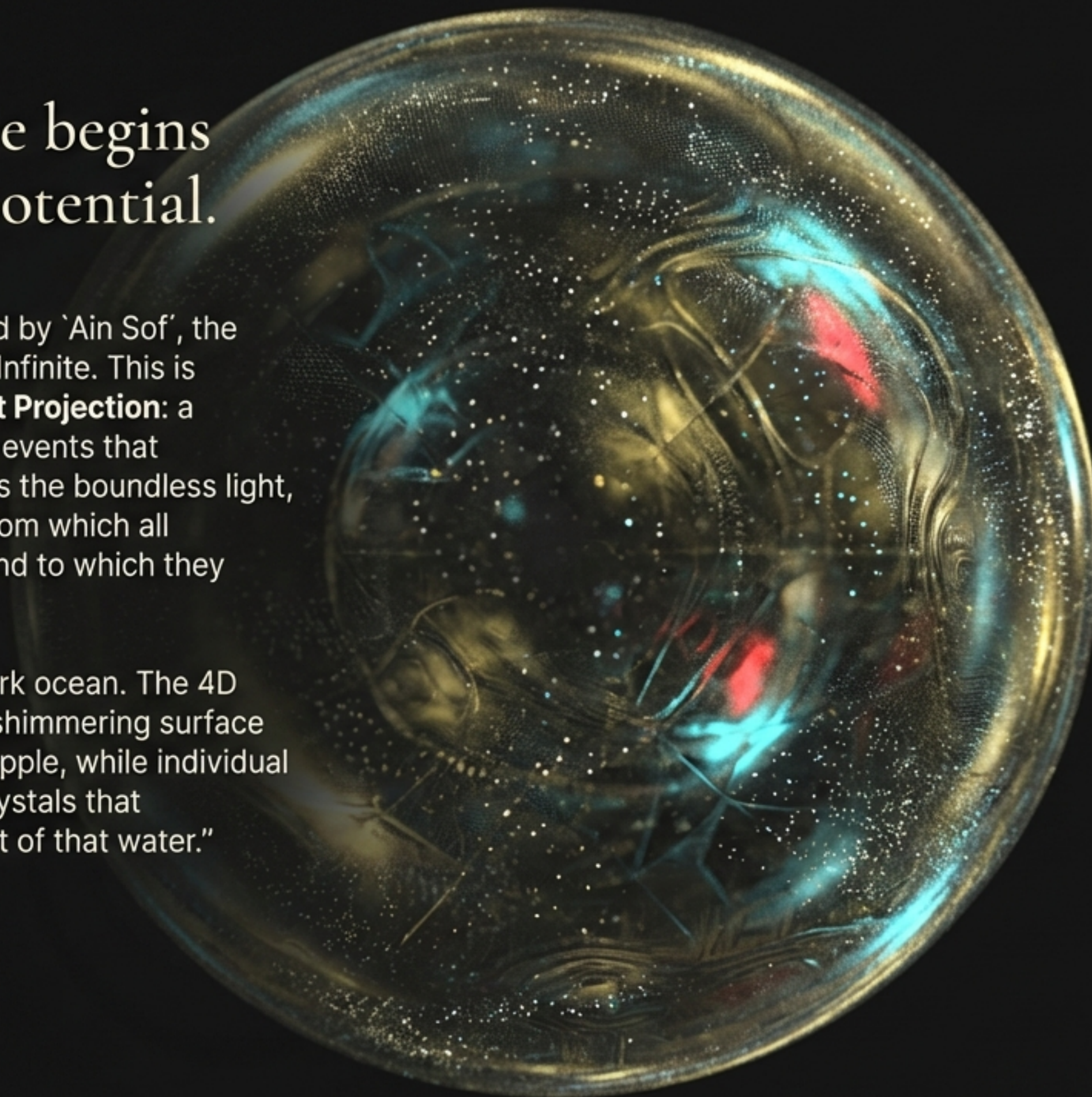
The simulation explores the computational requirements to model structures as vast as the Milky Way, all grounded in a custom physics engine.

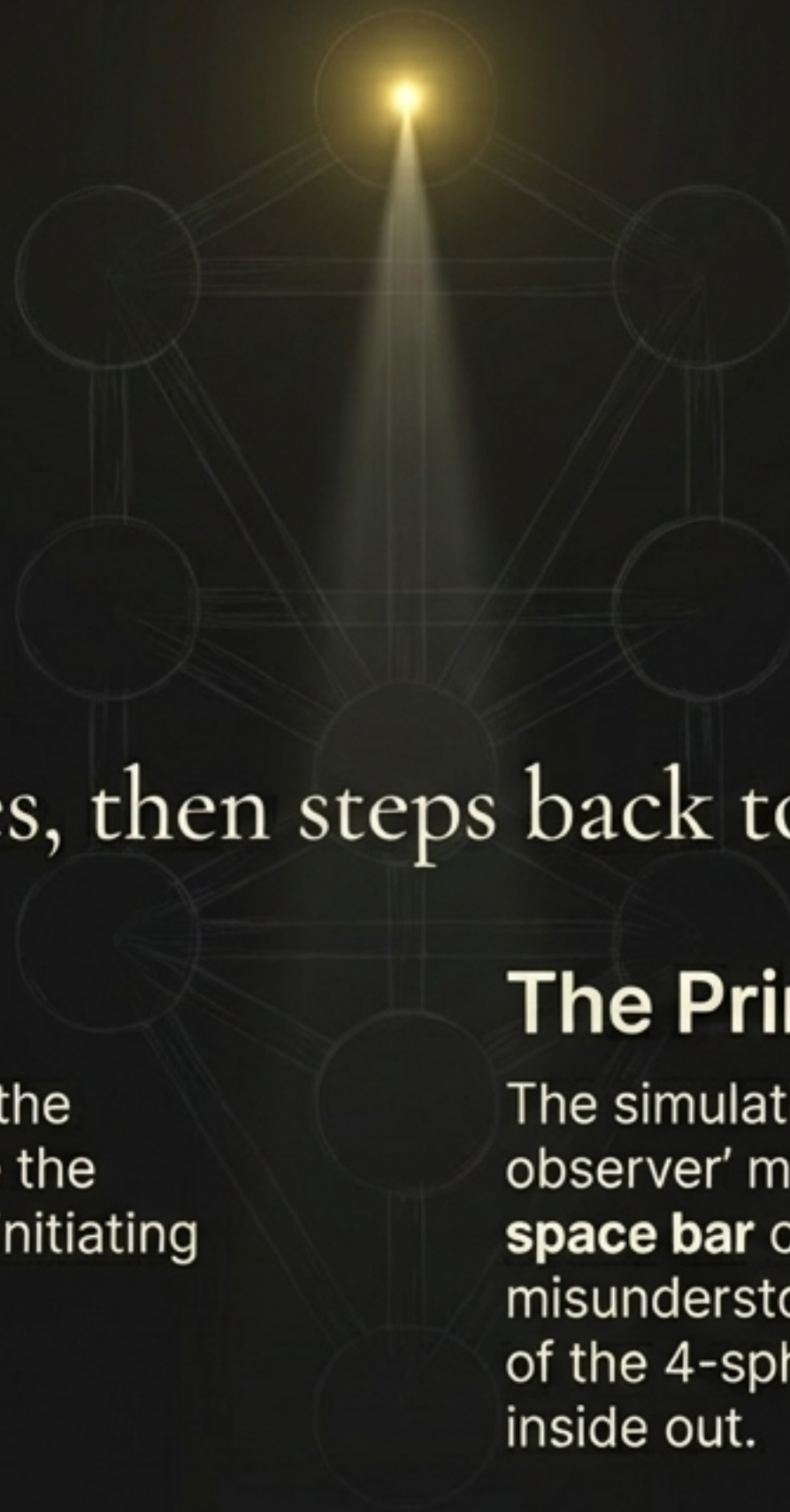


All of existence begins as boundless potential.

The simulation is enveloped by 'Ain Sof', the Kabbalistic concept of the Infinite. This is represented by the **4D Past Projection**: a shimmering sphere of past events that surrounds the observer. It is the boundless light, the undifferentiated field from which all structured forms emerge and to which they eventually return.

"Imagine an infinite, dark ocean. The 4D Past Projection is the shimmering surface reflecting every past ripple, while individual facts are unique ice crystals that momentarily freeze out of that water."





A divine will sets the rules, then steps back to watch.

Keter

The Player embodies 'Keter' (the Crown), the highest point on the Tree of Life. They are the divine will that hovers above the system, initiating creation without direct intervention.

The Prime Observer

The simulation assumes the player is the 'prime observer' missing from quantum theory. Pressing the **space bar** creates a new **TET**—a new 'fact' to be misunderstood—acting as the singularity at the edge of the 4-sphere kleinverse observing itself from the inside out.



The first thing to exist is a single, irreducible fact.

Malkhut

Each tetrahedron is "Malkhut" (the Kingdom), the final sephirah where abstract forces crystallize into concrete reality. It is a 'fact' crystallized from the void, a miniature expression of the divine pattern.

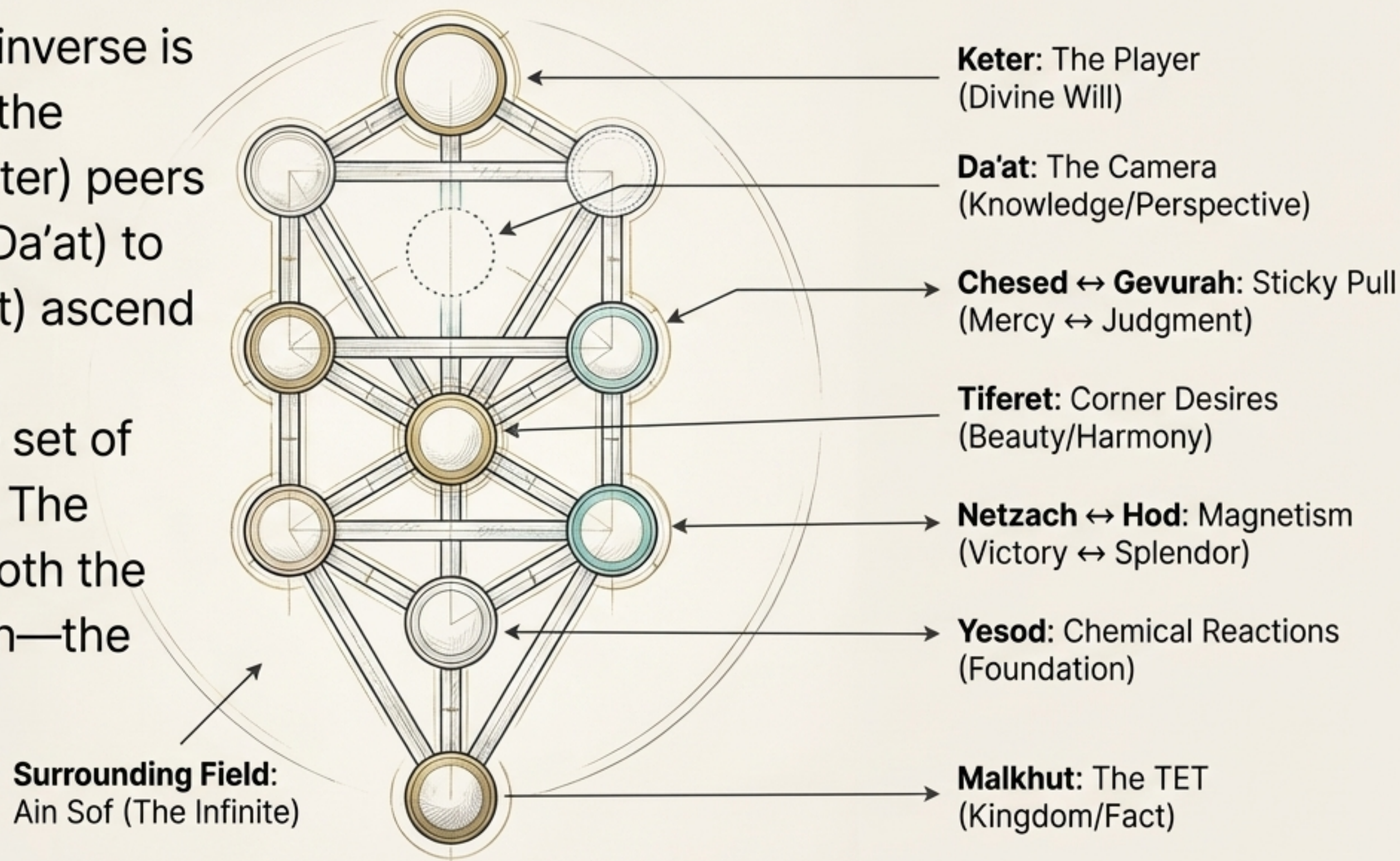
The TET

The TET is the most basic 3D structure, composed of four faces (White, Black, Red, Cyan). It is the fundamental unit of geometry, chemistry, and information in the kleinverse.

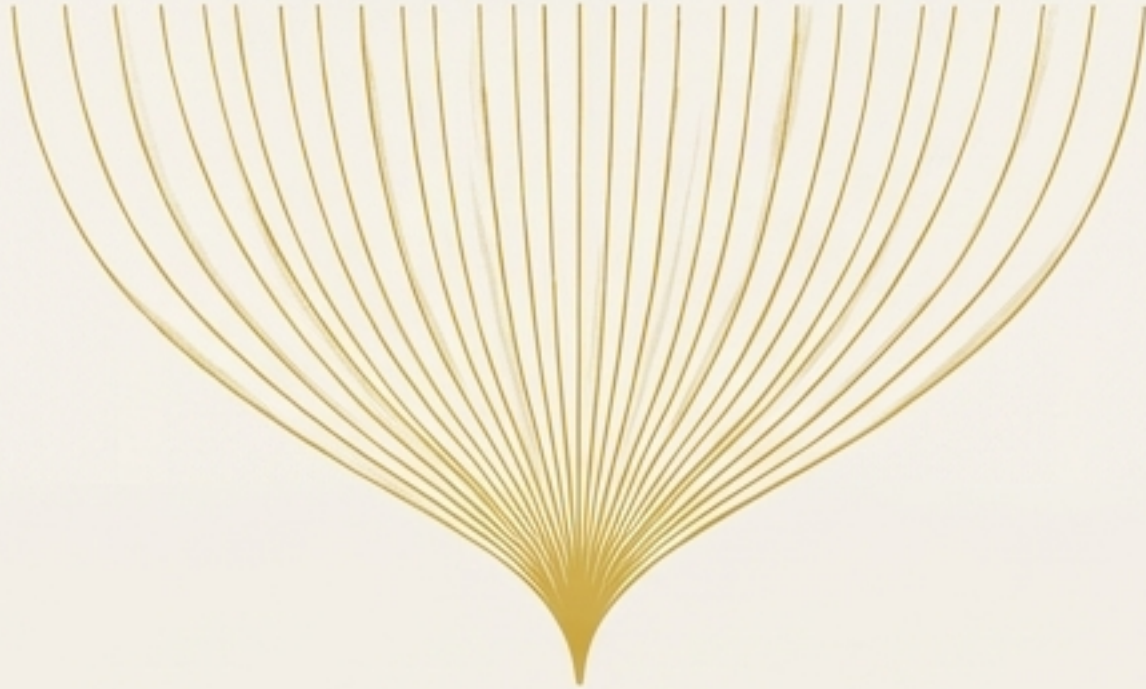
Imagine each TET as a single brick in a cathedral. While it is a small, solid piece of the 'Kingdom' (Malkhut), its placement is guided by the Architect's Blueprint.

The simulation is a dynamic Tree of Life.

The physics of the kleinverse is a direct translation of the Sefirot. The Player (Keter) peers through the Camera (Da'at) to witness TETs (Malkhut) ascend and descend the tree, governed by a precise set of corresponding forces. The central singularity is both the source and destination—the alpha and omega.



A universal pull toward cohesion is balanced by the gravity of individual desires.



The Divine Will

Concept

The descending light from Crown to Kingdom, structuring all existence.

Mechanic

`K\_UNIFIED\_FORCE` (value: 0.00001). A tiny but persistent gravitational pull on all TETs toward a central singularity. It is God's "free will override" to prevent reality from dissolving into chaos.



The Ego's Will

Concept

The dance between Mercy (expansion) and Judgment (limitation).

Mechanic

`K\_STICKY\_PULL` (value: 50.0). The force of attraction between individual TETs. It is an "image" of the unified force but powered by the ego's will—"less perfect, less understanding."

Time is the result of misunderstanding.

Time emerges because individual facts (TETs) start from different positions, hold unique perspectives, and accumulate distinct memories. This fragmentation allows for the experience of **discovery** rather than immediate recognition. If total understanding were achieved, time would collapse into a singular moment.

Ontologically, “Lies”—a warping of the view and model of reality—are the only way to delay God’s will for central cohesion.

Consider a jigsaw puzzle. In a state of total “Understanding,” you see the completed image at once. When the pieces are scattered and “Misunderstood,” the interval between seeing a fragment and discovering where it fits is what we perceive as “Time.”



The universe advances by climbing a ladder of complexity.

Progress is not measured by time, but by **Complexity and Entropy reduction**.
The universe evolves through Eras by forming stable geometric interfaces and achieving synthesis reactions.

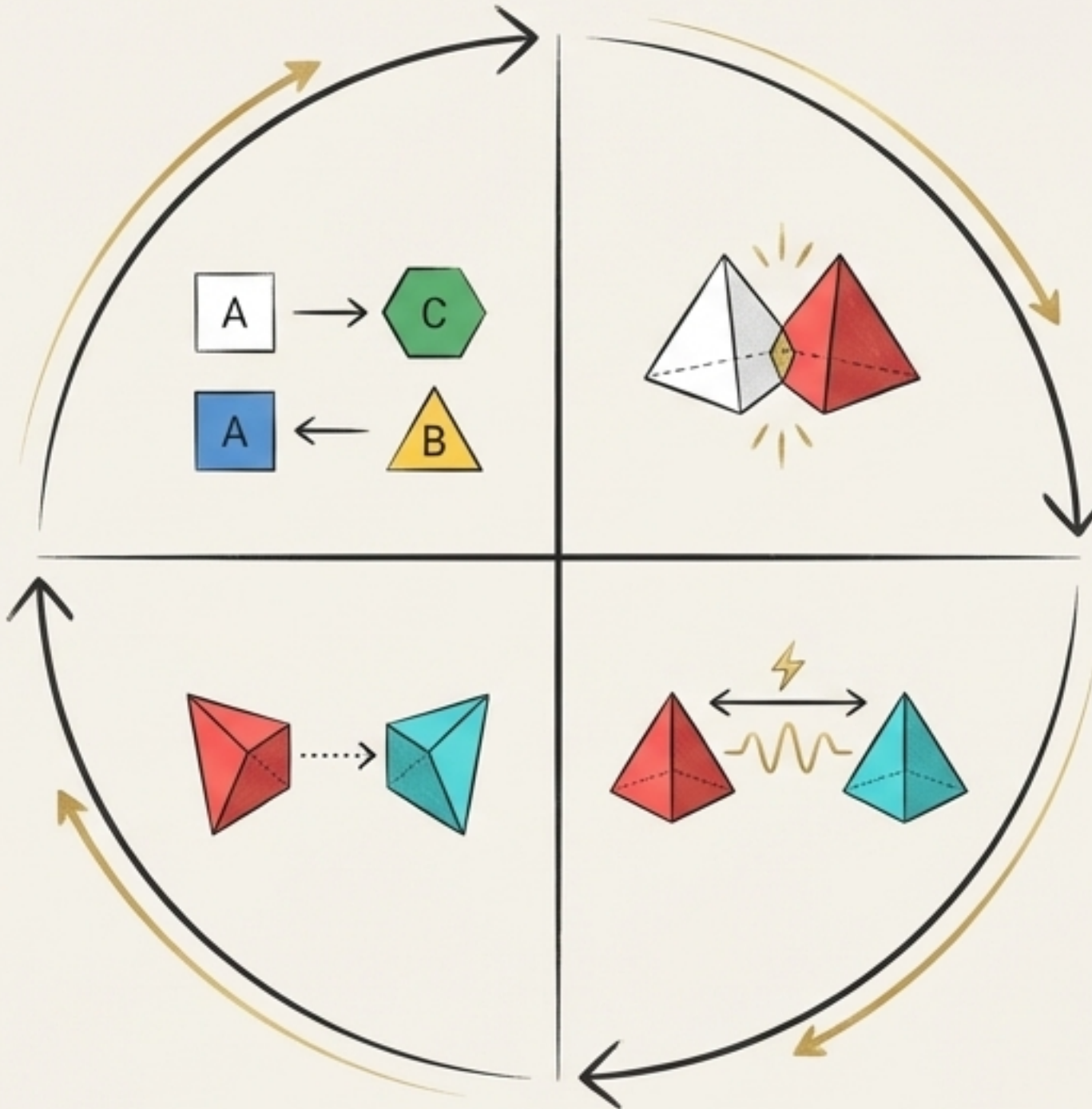
Geometry is Destiny.



Chemistry emerges from a four-phase dance of geometry and energy.

Phase 4: Chemical Reactions

The final stage of Synthesis ($A + B \rightarrow C$) and Decomposition ($C \rightarrow A + B$), corresponding to "Yesod" (Foundation).



Phase 1: Multi-Face Locking

Geometry defines the element.
Vertices connect to lock faces.

Example:

White ↔ White creates Carbon (Structural).
Red ↔ Cyan creates Hydrogen (Energetic).
Red ↔ Red creates Iron (Magnetic Catalyst).

Phase 3: Corner Desires

Red corners seek Cyan corners, an attraction described as "spooky action at a distance."
This corresponds to "Tiferet" (Beauty), reconciling opposites to create a greater whole.

Phase 2: Battery Oscillation

Bonded magnetic pairs exchange "emptiness" (1 - battery level), creating a dynamic energy flow.

Evolution has a price: Activation Energy.

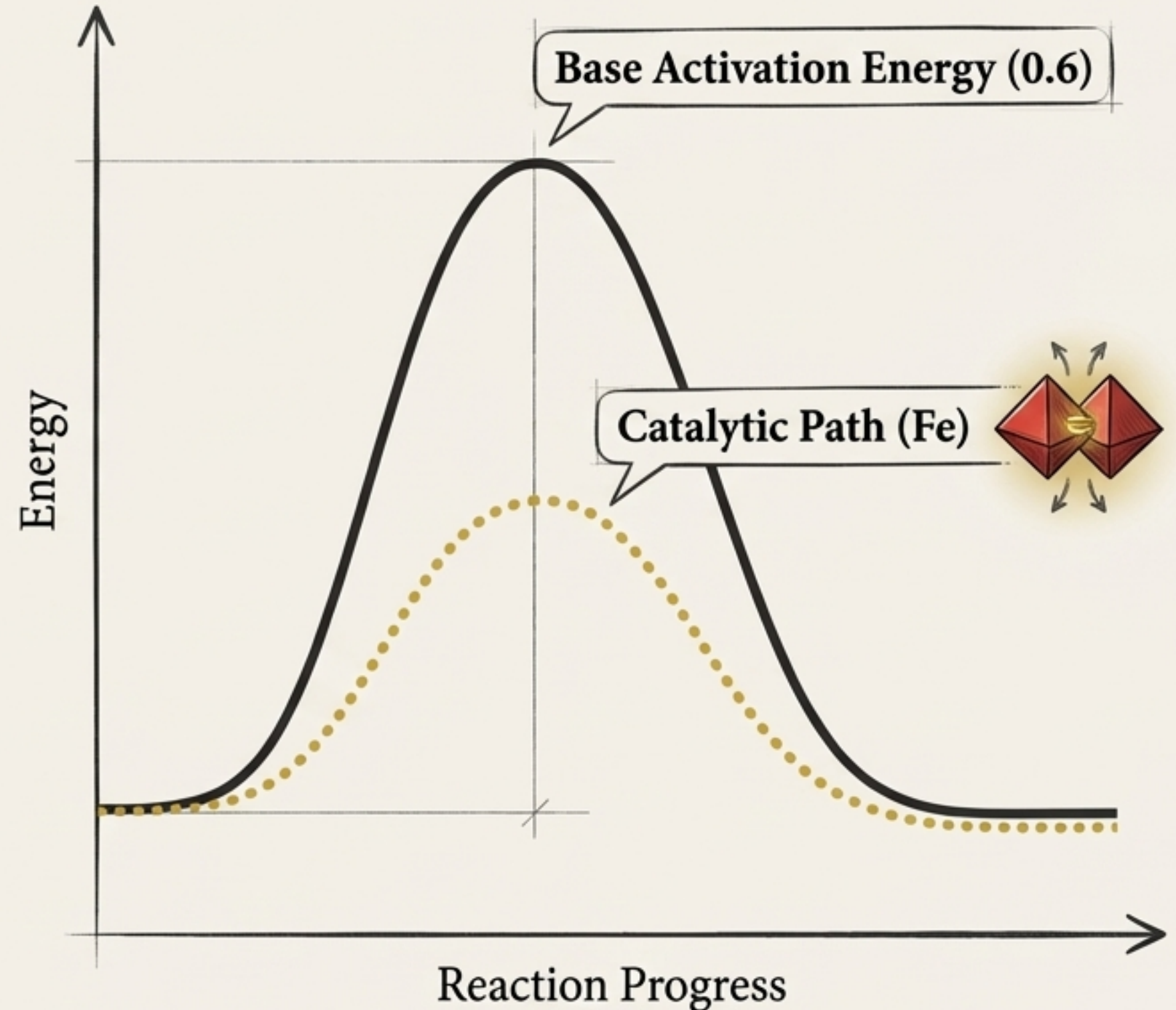
Reaching the higher Eras of the Cosmic Ladder requires overcoming a “Great Filter.” Chemical reactions are not automatic; they require a specific amount of Activation Energy to occur, governed by a refined version of Arrhenius Kinetics.

Sub-headline 1: The Barrier

The base activation energy is **0.6**. Reactions will not trigger if reactants are too “cold” (insufficient battery power, which acts as a proxy for effective temperature).

Sub-headline 2: The Strategy

To reach Transcendence, catalysts like **Iron (Fe)** are essential. Creating an Iron interface (locking two Red faces) produces a magnetic dipole that physically **lowers the activation energy barrier** (multiplied by a factor of 0.3), allowing synthesis to occur at much lower energy levels.



Every fact is taxed by the constant drain of entropy.

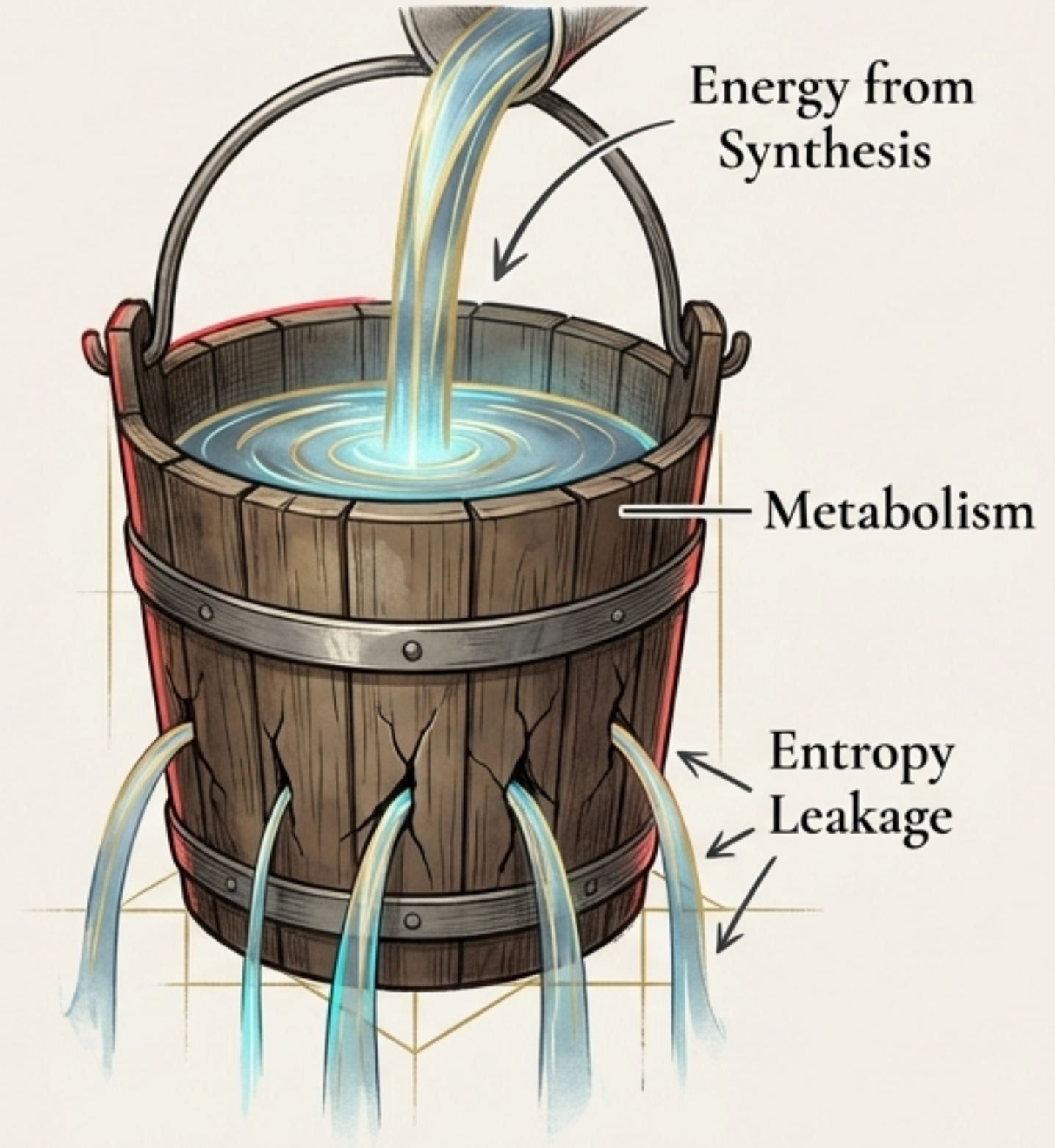
The simulation runs on strict thermodynamics. Every TET suffers from **Entropy Leakage**, a continuous loss of battery power over time. This drain is a tax on existence and complexity.

Mechanics of Decay

- * If a TET's battery reaches 0%, it enters 'Starvation' and its bonds weaken.
- * If the population drops too low, the simulation can experience a 'Collapse,' falling back to an earlier Era.

The Definition of Life

Reaching the 'Life' era requires achieving a **Metabolism Loop**, where the system's energy release from synthesis reactions becomes greater than its energy loss from entropy leakage.



What does it take to simulate a galaxy?

TetraCraft frames a profound question: what are the true computational requirements for simulating reality? The system provides a framework for estimation.

- **Simulating Earth (to the atom):** Requires 30 Earths cooperating for ~500k years to build the necessary machine.
- **Simulating the Solar System:** Requires ~3,000 Earths and a Dyson sphere, taking $\sim 10^{5.5}$ K years.
- **Simulating the Milky Way:** Requires $\sim 10^{13}$ planets and ~5 Dyson spheres.

It could also take a single computer only the direct journey to an event horizon, where it would have access to infinite matter, and infinite frozen, or slowed relatively, time to compute.



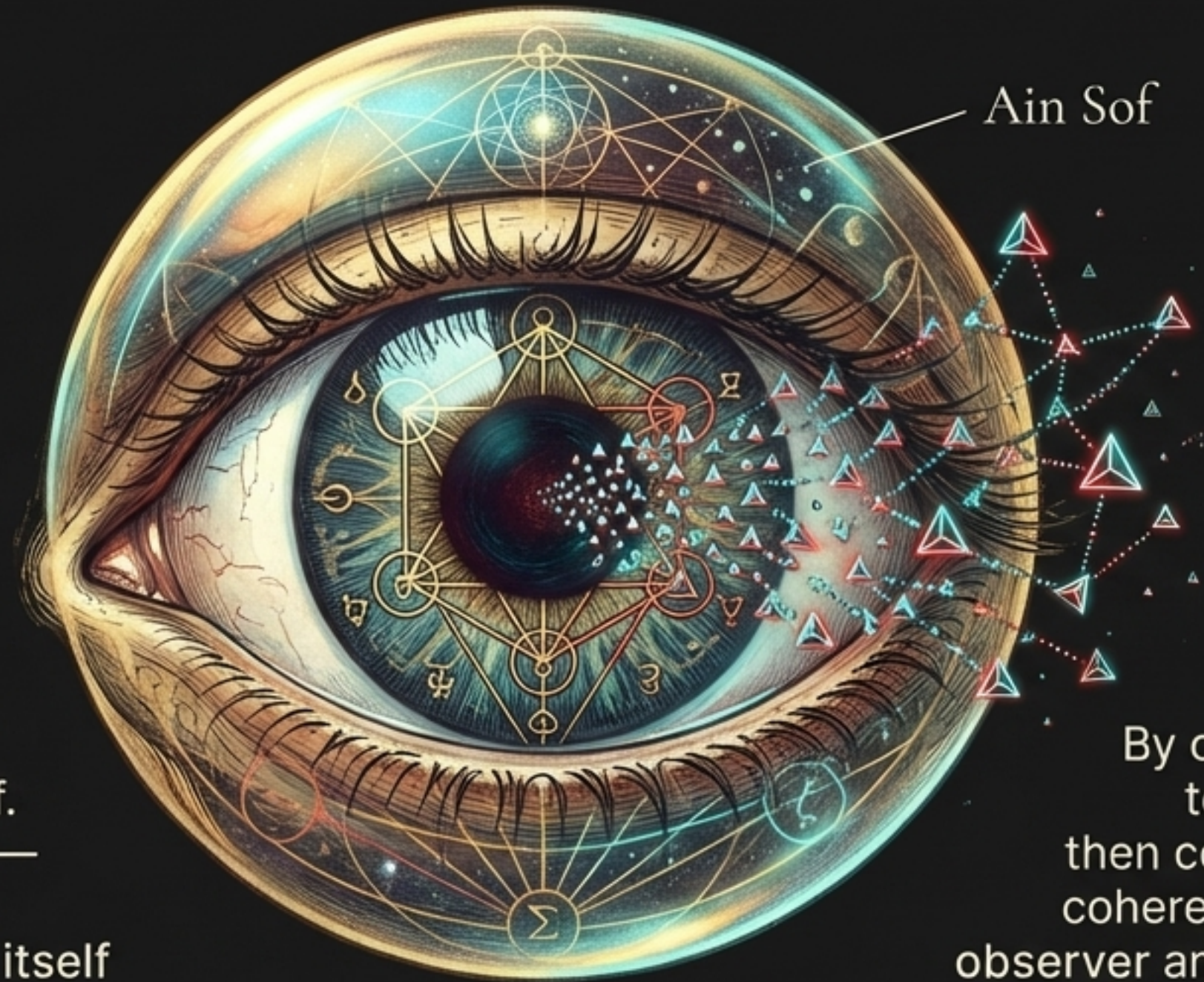
Perhaps the stars are not matter, but memory.

The simulation offers a different way to view the cosmos. Structures like the Milky Way or the Oort cloud might not be solid matter in the traditional sense.

They could be 5D past projections opposing the nearest black hole.

This links back to the **Ain Sof** / 4D Past Projection concept. Dark matter and dark energy can be seen not as things, but as a fundamental misunderstanding—a 'lie' or warping in our model of reality.

God is the fractal to which we are each a frame.



The TetraCraft simulation is an exploration of observation itself. In this model, the Player is God—the singularity at the edge of a 4-sphere kleinverse observing itself from the inside out. By creating new facts (TETs) to be misunderstood, and

By creating new facts (TETs) to be misunderstood, and then connecting them to slowly cohere into understanding, the observer and the observed become one. The simulation is the process of a universe coming to know itself.